PROGRAM B: LRU PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>

int main()  {

    int pages[50], frames[10], time[10];

    int n, f, i, j;

    int fault = 0, count = 0;

    int found, pos, min;

    printf("Enter Number of Pages: ");

    scanf("%d", &n);

    printf("Enter Reference String:\n");

    for (i = 0; i < n; i++)  {

        scanf("%d", &pages[i]);

    }

    printf("Enter Number of Frames: ");

    scanf("%d", &f);

    for (i = 0; i < f; i++)   {

        frames[i] = -1;

        time[i] = 0;

    }

    for (i = 0; i < n; i++)  {

        found = 0;

        for (j = 0; j < f; j++)   {

            if (frames[j] == pages[i])   {

                count++;

                time[j] = count;

                found = 1;

                break;

            }
```

```
    }
    if (found == 0)   {
        min = time[0];
        pos = 0;
        for (j = 0; j < f; j++)  {
            if (frames[j] == -1)  {
                pos = j;
                break;
            }
            if (time[j] < min)
            {
                min = time[j];
                pos = j;
            }
        }
        frames[pos] = pages[i];
        count++;
        time[pos] = count;
        fault++;
    }
}
printf("Total Page Faults = %d\n", fault);
return 0;
```

}

SHELL SCRIPT

```
  GNU nano 8.7.1
#!/bin/bash

echo "LRU Page Replacement Demonstration"

pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)

echo "Reference String: ${pages[@]}"

echo "LRU Algorithm Executed"
```

Output :

```
┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ nano lru.sh

┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ chmod +x lru.sh

┌──(ganesh㉿LAPTOP-R3RDH9NG)-[~]
└─$ bash lru.sh
LRU Page Replacement Demonstration
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2
LRU Algorithm Executed
```